NON-CALCULATOR – 30 MINUTES

SAMPLE ASSESSMENT MATERIAL

Do not open this paper until you are told to do so by the invigilator.

Overall assessment marks available: **60**
Overall assessment time limit: **2 HOURS**

There are **TWO** Sections to this assessment:

- **Section A** includes Task 1. You **must not** use a calculator for this section.

  Total marks available: 15. Time limit: 30 minutes
- **Section B** includes Task 2, 3 and 4. You can use a non-scientific calculator for this section

  Total marks available: 45. Time limit: 1 hour and 30 minutes

For Section A you need:

- This question and answer paper
- A pen with black or blue ink
- A pencil
- A ruler

INTERNET ACCESS IS NOT PERMITTED AND YOU MUST NOT USE A CALCULATOR

The invigilator will stop the assessment after 30 minutes. You must hand in this question and answer paper at this point.

The invigilator will then hand out **Section B** and a non-scientific calculator. You will then have a further 1 hour and 30 minutes to complete **Section B**.

Instructions

1. Please sign and date below to confirm that your details are correct and that you have understood the instructions.
2. Read each task and question carefully.
3. Remember to show all your workin
4. The number of marks available for rks to guide you on how long to spend on ea
5. Answer **all** questions using the spa
6. If you have time, check your work f this question and answer paper, you will n
7. If you use extra paper, write your n are answering on it and securely attach it

Learner name:	Joe Bloggs
Learner number:	10000000
Centre number:	10000
Signature:	
Today's date:	

OFFICIAL MFS MODEL SOLUTION

60 / 60 Marks (100% Pass)

· Assessment: Sample Assessment Materials (SAM)
· Complete step-by-step worked models
· Official Mark Scheme Allocations [MI / AI]
· Exam Board: Open Awards (Ofqual Regulated)

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Section A

Task 1 (15 marks)

Question 1

The table below shows the distances walked by Jay each day in a week.

Day of week	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
Distance walked	$1\frac{1}{2}$ miles	$\frac{3}{4}$ mile	$1\frac{1}{2}$ mile	$\frac{1}{3}$ mile	$\frac{1}{3}$ mile	$\frac{3}{4}$ mile	0 miles

Step 1 Convert to common denominator 12 and add:

1 1/2 = 1 6/12 3/4 = 9/12 1 1/2 = 1 11/12

Sum = (1 + 1) + (6/12 + 9/12 + 11/12) = 2 + 26/12

2 + 2 2/12 = 4 2/12 = 4 1/6 miles (or 4.17 miles)

[M1]

What was the total distance Jay walked over the whole week? (2 marks)

[A1]

Show your calculations and/or workings out here.

4 1/6 miles (or 4.17 miles)

Write your answer in this box.

Question 2

Put these fractions in order of size, smallest to largest:

$\frac{4}{3}$ $\frac{3}{4}$ $\frac{3}{8}$ $\frac{5}{8}$ $\frac{7}{6}$

(1 mark)

Convert all fractions to common denominator 24:

3/8 = 9/24 | 5/8 = 15/24 | 3/4 = 18/24 | 7/6 = 28/24 | 4/3 = 32/24 [1 mark]

Show your calculations and/or workings out here:

3/8, 5/8, 3/4, 7/6, 4/3 ✓

Long division: 273,696 ÷ 24:

273696 ÷ 24 = 11,404

Write your answer in this box. [1 mark]

11,404 ✓

Question 3

Calculate 273696 ÷ 24. (1 mark)

Show your calculations and/or workings out here:

Write your answer in this box.

Question 4

Calculate the surface area of a cube when the length of a side a = 15cm. (2 marks)

Surface area = $6a^2$

Show your calculations and/or workings out here:

Step 1: Calculate area of one square face (15 cm × 15 cm):

Face area = 15 × 15 = 225 cm² [M1]

Step 2: A cube has 6 identical square faces. Total surface area:

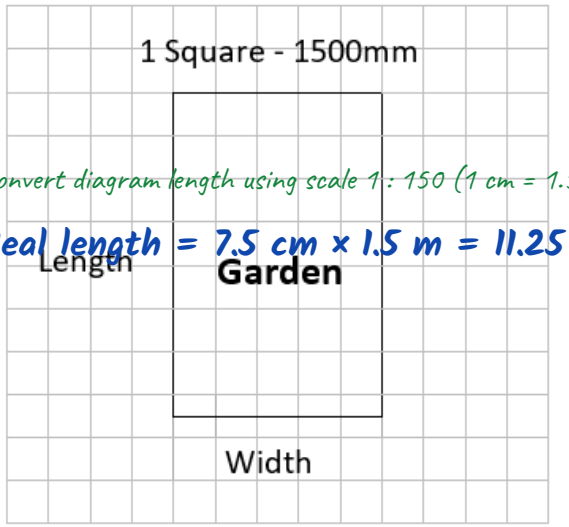
Total surface area = 6 × 225 = 1350 cm² [M1 A1]

Write your answer in this box.

1350 cm²

Question 5

a) Simon is redesigning his garden. He has drawn his garden on the diagram below where 1 square = 1500mm.



Convert diagram length using scale 1 : 150 (1 cm = 1.5 m):

Real length = 7.5 cm × 1.5 m = 11.25 metres [M1 A1]

11.25 m

Using the grid, calculate the actual length of the garden in metres? (2 marks)

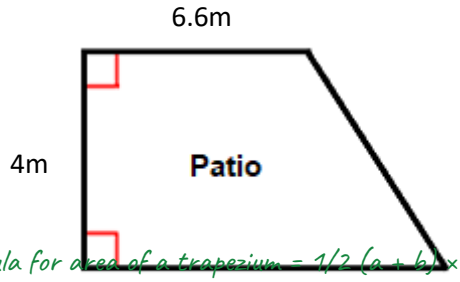
Show your calculations and/or workings out here:

Write your answer in this box.

b) Simon is planning to build a patio in his garden. The patio will have four sides.

He has drawn a sketch of the patio below.

Sketch not drawn to scale



Step 1: Formula for area of a trapezium = 1/2 (a + b) × h:

Parallel sides a = 8.4m, b = 6.6m; height h = 4.0m

Area = 1/2 × (8.4 + 6.6) × 4 = 1/2 × 15.0 × 4 [M1]

Calculate the area of the patio. (3 marks)

Area = 7.5 × 4 = 30 m² [M1 A1]

Show your calculations and/or workings out here:

30 m²

Write your answer in this box.

c) For the foundation of the patio, Simon will use a dry mixture of sand and cement.

He will need 20kg of mixture for each square metre of patio.

To make the mixture he needs to mix sand and cement in the ratio of 5:1.

Calculate how many 25kg bags of cement he will need. (4 marks)

Show your calculations and/or workings out here:

Step 1: Total dry mixture needed for 30 m^2 (20 kg per m^2):

Total mixture = 30 × 20 kg = 600 kg [M1]

Step 2: Ratio of cement to sand is 1 : 5 (Total = 1 + 5 = 6 parts):

Cement required = 600 kg ÷ 6 = 100 kg [M1]

Step 3: Number of 25 kg bags of cement: 100 kg ÷ 25 kg = 4 bags [M1 A1]

4 bags of cement ✓

Write your answer in this box.

[End of Section A]

LEVEL 2 FUNCTIONAL SKILLS QUALIFICATION IN MATHEMATICS

SECTION B - QUESTION AND ANSWER PAPER (RFSML2SAM01)

CALCULATOR – 1 HOUR AND 30 MINUTES

SAMPLE ASSESSMENT MATERIAL

Do not open this paper until you are told to do so by the invigilator.

Overall assessment marks available: **60**
Overall assessment time limit: **2 HOURS**

There are **TWO** Sections to this assessment:

- **Section A** – please ensure you have handed in Section A before beginning Section B
- **Section B** includes Task 2, 3 and 4. You can use a non-scientific calculator for this section.

  Total marks available: 45. Time limit: 1 hour and 30 minutes.

For Section B you need:

- This question and answer paper
- A pen with black or blue ink
- A pencil
- A ruler
- A non-scientific calculator

INTERNET ACCESS IS NOT PERMITTED
You now have a further 1 hour and 30 minutes to complete **Section B**.

Instructions
1. Please sign and date below to confirm that your details are correct and that you have understood the instructions.
2. Read each task and question carefully.
3. Remember to show all your workings out clearly.
4. The number of marks available for each question is shown in brackets. Use these marks to guide you on how long to spend on each question.
5. Answer **all** questions using the space provided on this question and answer paper.
6. If you have time, check your work for **Section B** at the end.
7. If you use extra paper, write your name, learner number and the question number you are answering on it, and securely attach it to this question and answer paper.
8. At the end of this section (**Section B**), hand in this question and answer paper and all notes to the invigilator.

Learner name:	Joe Bloggs
Learner number:	10000000
Centre number:	10000
Signature:	
Today's date:	

Section B

Task 2 (15 marks)

Question 6

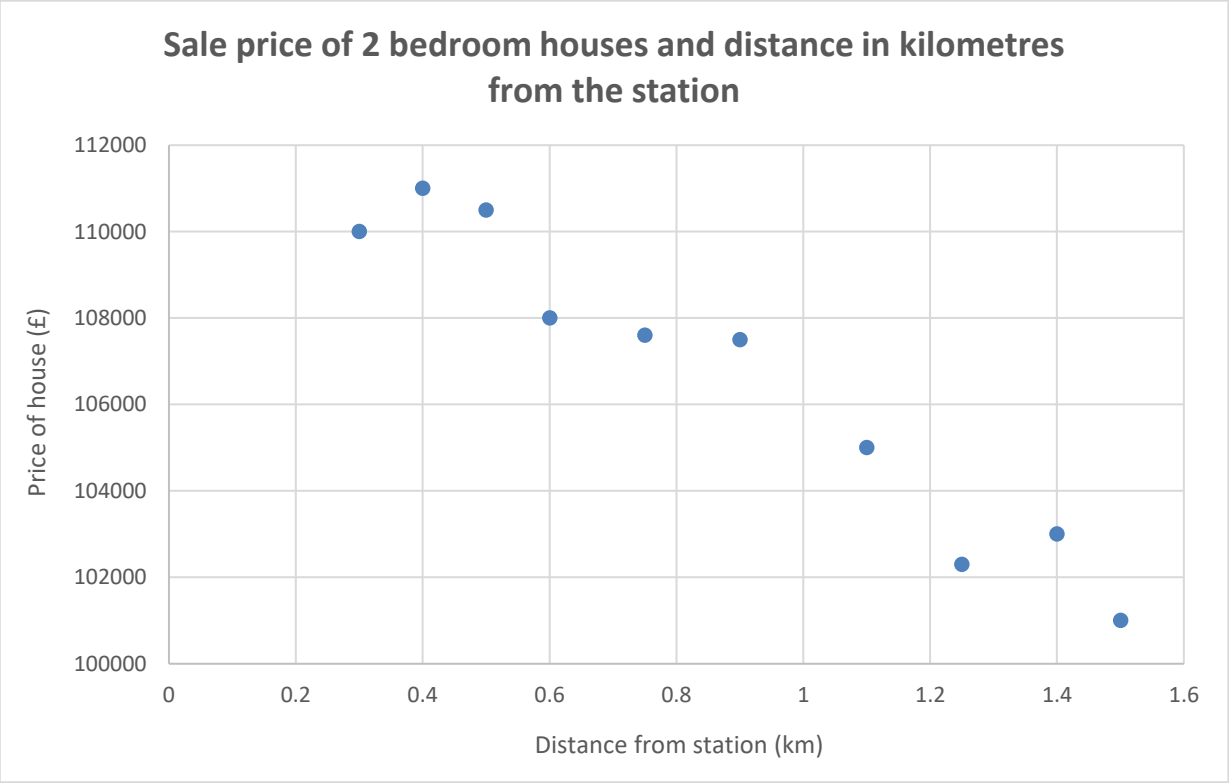
Khalid wants to buy a two-bedroom house no further than 0.6 miles from the station.

Khalid has saved a deposit of £4 875. He can afford a mortgage of 3.5 times his earnings which is £28 145 per annum.

The scatter graph shows information about the price and distance from the station of recent two-bedroom house sales in the area.

Can Khalid afford to buy a two-bedroom house within 0.6 miles of the station? Give a reason for your answer. (5 marks)

1 mile = 1.6 km



See full calculations and comparison on top of next page -->

Show your calculations and/or workings out here:

Step 1: Calculate total maximum house budget:

Budget = (3.5 × £28,145) + £4,875 = £98,507.50 + £4,875 = £103,382.50 [M1]

Step 2: Convert maximum distance 0.6 miles to km (1 mile = 1.6 km):

0.6 miles × 1.6 = 0.96 km (under 1.0 km from station) [M1]

Step 3: From scatter graph, houses within 0.96 km cost at least £106,000.

£103,382.50 < £106,000, so he cannot afford a house at this distance [M1 A1]

No, budget £103,382.50 is less than required ~£106,000

Write your answer in this box, giving a reason for your answer.

Question 7

Find the mode in the following set of numbers. (1 mark)

8 8.5 8 7 11 23 9 11 7.5 11 7

Mode is the most frequently occurring value in the list

11 appears 3 times (more than any other number) **[1 mark]**

Write your answer in this box.

11

Question 8

Calculate the median of the following set of numbers. (2 marks)

10 10.5 11 12 15 23 9 9.5

Show your calculations and/or workings out here:

Step 1: Order numbers from smallest to largest (8 values):

9, 9.5, 10, 10.5, 11, 12, 15, 23 **[M1]**

Step 2: Median is mean of 4th and 5th middle values:

Median = (10.5 + 11) ÷ 2 = 21.5 ÷ 2 = 10.75 **[A1]**

Write your answer in this box.

10.75

Question 9

Amy wants to catch the 10.12am train from Darlington to Chesterfield.
She needs to allow 10 mins to buy a ticket and get to the platform.
She lives 2 miles from the station and knows that she can walk at 3mph.
At what time should she leave home? (2 marks)

Show your calculations and/or workings out in the space below:

Step 1: Walking time at 3 mph for 2 miles:

Time = 2 ÷ 3 = 2/3 hour = 40 minutes [M1]

Step 2: Total time needed before train departure at 10:12 am:

40 mins walking + 10 mins at platform = 50 minutes total

Count back 50 mins from 10:12 am: 10:12 - 50 mins = 09:22 am [A1]

Write your answer in this box.

09:22 am

Question 10

Tom is given £8 500 to go towards a deposit to buy his first house.

Tom sees these two savings accounts.

Money saver account	Bonus saver account
1.75% per year. To be added at the end of each year.	Save for 3 years and receive a single bonus of 5.25%

Tom puts his money in the Money saver account.

How much more money will Tom have after 3 years compared to the Bonus Saver account?

(5 marks)

Show your calculations and/or workings out here:

Step 1: Money Saver account compound interest (1.75% per year):

- Year 1: £8500 × 1.0175 = £8648.75
- Year 2: £8648.75 × 1.0175 = £8800.10 [M1 A1]
- Year 3: £8800.10 × 1.0175 = £8954.10

Step 2: Bonus Saver account (single 5.25% bonus after 3 years):

£8500 × 1.0525 = £8946.25 [M1]

Step 3: Difference = £8954.10 - £8946.25 = £7.85 more [A1]

Write your answer in the box.

£7.85 more

Task 3 (15 marks)

Question 11

The formula below is used to calculate the percentage fuel saving when driving at a reduced speed compared to a higher speed.

$$F = 100 \times \left(\frac{a-b}{b}\right)^2$$

F = % fuel savings
a = original average speed
b = reduced average speed

Calculate F when the speed is reduced from 60 mph to 50 mph. (3 marks)

Show your calculations and/or workings out below:

Step 1: Substitute values a = 60 mph, b = 50 mph into formula:

$F = 100 \times ((a - b) / b)^2 = 100 \times ((60 - 50) / 50)^2$ [M1]

Step 2: Evaluate bracket: $(10 / 50)^2 = (0.2)^2 = 0.04$ [M1]

Step 3: Multiply by 100: $F = 100 \times 0.04 = 4\%$ [A1]

Write your answer in this box.

4 %

Question 12

a) Raheema is concerned about the environment and is looking for ways to be more eco-friendly.

Raheema is researching the use of solar panels for her house. She has found some information on the total number of sun hours per month where she lives for 2016 and 2017.

Total sun hours per month		
	2016	2017
January	21	47
February	75	61
March	112	119
April	147	128
May	206	214
June	143	108
July	112	144
August	146	126
September	105	94
October	97	56
November	64	6
December	21	

Average sun hours per month 2017	
Mean	94.5

Raheema thinks the total number of sun hours was higher in December 2017 than in December 2016. Is she correct? (3 marks)

Show your calculations and/or workings out here:

Step 1: Total sun hours in 2017 from mean = 94.5 × 12 = 1134 hours [M1]

Step 2: Sum Jan–Nov = 47+61+119+128+214+108+144+126+94+56+6 = 1103 [M1]

Step 3: Dec 2017 = 1134 − 1103 = 31 hours. Raheema is correct. [A1]

Yes, Raheema is correct (31 hours in December)

Write your answer in this box.

b) Which year had the greatest range of sun hours? (1 mark)

Show your calculations here:

Compare range (Max - Min) for both years:

- 2016 Range = 206 - 21 = 185 hours
- 2017 Range = 214 - 6 = 208 hours (Greatest range) [1 mark]

2017 (range of 208 hours vs 185 hours)

Write your answer in this box.

c) To generate the maximum amount of electricity, a solar panel needs to face south and have a tilt angle of 30º. This will generate a maximum of 1.225kWh of electricity for each hour of sunshine. In June there were 108 hours of sunshine.

Raheema's roof faces south-west and has a tilt angle of 50º. To find out how much electricity her solar panel will produce, she needs to divide the maximum electricity that could be generated by a factor given in the table below.

Raheema usually pays £0.143 per kWh of electricity.

Tilt Angle	Facing South-west	Facing South	Facing South-east
60°	1.15	1.07	1.15
50°	1.09	1.03	1.08
40°	1.05	1.01	1.05
30°	1.04	1	1.04

How much would the electricity generated in June cost if she had to pay for it?

(3 marks)

Show your calculations and/or workings out here:

Step 1: South-west facing, 50° tilt factor = 1.09:

kWh generated per sun hour = 1.225 ÷ 1.09 = 1.12385 kWh [M1]

Step 2: In June 2017, sun hours = 108 hours:

Total kWh generated = 1.12385 × 108 = 121.376 kWh [M1]

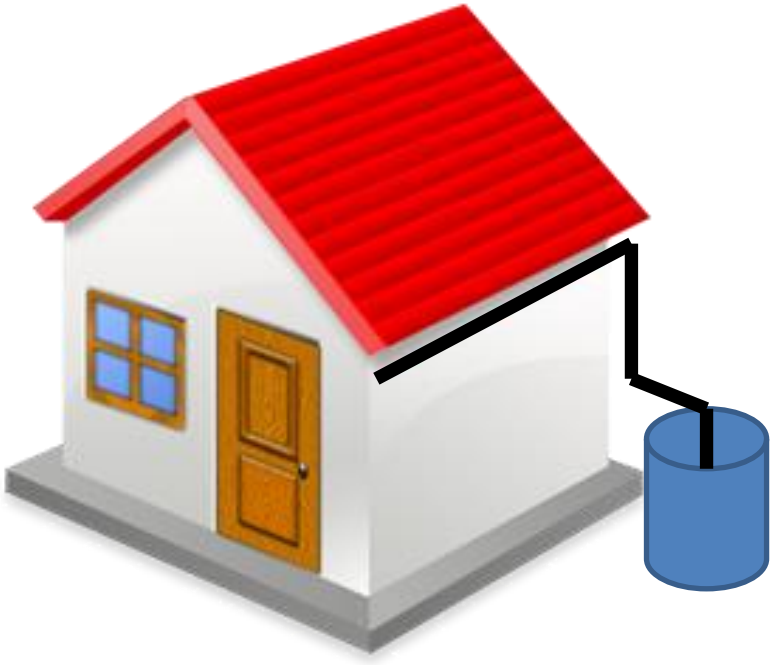
Step 3: Cost at £0.143 per kWh = 121.376 × £0.143 = £17.356... --> £17.36 [A1]

£17.36

Write your answer in this box.

d) Raheema finds that she can be more environmentally friendly by collecting rain water from her drain pipe, so she can use it to water her garden.

Raheema buys a cylindrical container that is 80cm in diameter and 1 metre high.



Raheema thinks the container will hold at least 100 gallons of water. Is she correct?

(5 marks)

π = *3.14*

1m³ = 219.97 gallons

Show your calculations and/or workings out here:

Step 1: Radius = 80 cm ÷ 2 = 40 cm = 0.4 m; height = 1.0 m

Volume = π × r² × h = 3.14 × (0.4)² × 1.0 = 3.14 × 0.16 = 0.5024 m³ [M1 A1]

Step 2: Convert m^3 to gallons (1 m^3 = 219.97 gallons):

Capacity = 0.5024 × 219.97 = 110.51 gallons [M1 A1]

Explain your answer in this box.

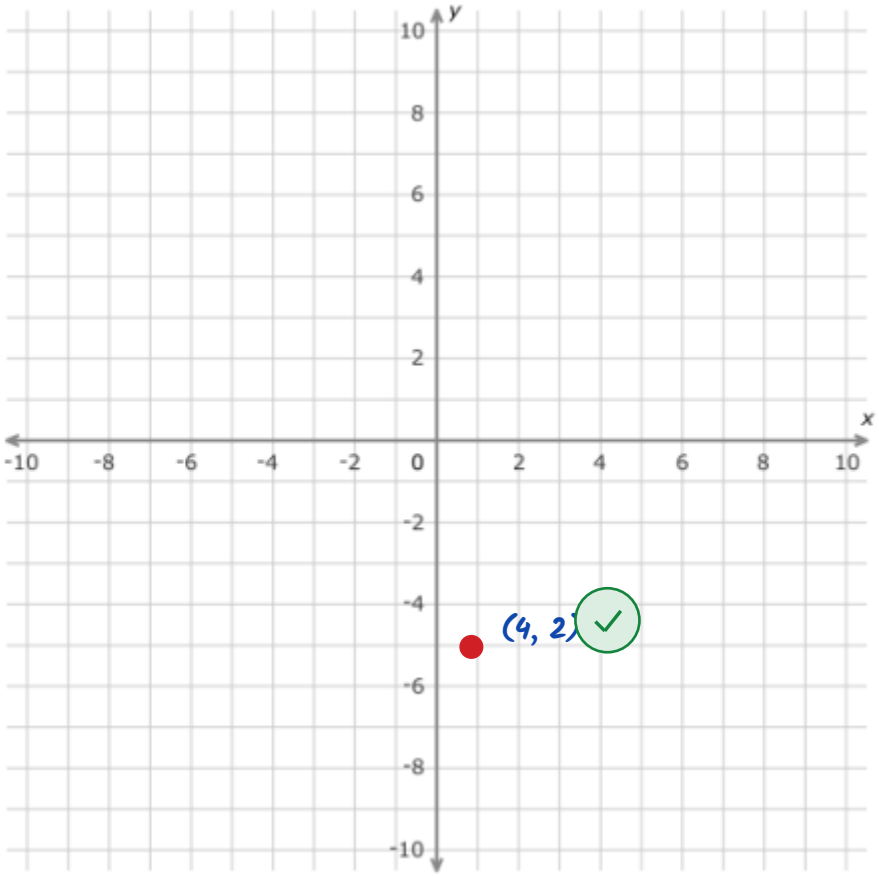
Explanation and Decision:

Yes, Raheema is correct: container holds 110.51 gallons, which is greater than 100 gallons (110.51 > 100). [A1]

Yes, capacity is 110.51 gallons (> 100 gallons)

Task 4 (15 marks)

Question 13



On the grid mark the point (4, 2). (1 mark)

Coordinate (4, 2): 4 units right along x-axis, 2 units up along y-axis. [1 mark]

Question 14

Give 144 as a fraction of 240 in its simplest form. (2 marks)

Show your calculations and/or workings out here:

Simplify fraction by dividing numerator and denominator by common factors:

144 / 240 = (144 ÷ 24) / (240 ÷ 24) = 6 / 10 = 3 / 5 [M1 A1]

3 / 5

Write your answer in this box.

Question 15

a) Sarah helps to organise a family fun day charity event each year.

Last year, 120 people attended the event each paying a £2.50 entry fee.

	Money taken during the event (£)	**Percentage of money taken (%)**
Entry fees		
Cake stall		19%
Bouncy castle		32%
Tombola		9%
Wheel of fortune		15%

It cost £175 to hire the Village Hall for the event and a further £85 for prizes.

How much profit did Sarah make for charity? (4 marks)

Show your calculations and/or workings out here:

Step 1: Entry fees = 120 people × £2.50 = £300.00

Step 2: Other percentages = 19% + 32% + 9% + 15% = 75%

Entry fees percentage = 100% - 75% = 25% (1/4 of total) [M1]

Step 3: Total income = £300 × 4 = £1200.00 [M1]

Step 4: Profit = Total income - Costs = £1200 - £175 - £85 = £940.00 [A1]

£940.00

Write your answer in this box.

b) Sarah bakes 15 identical cakes for the charity event. Each cake is circular with a radius of 80mm.

She plans to decorate each cake with a piece of ribbon around its edge.

She wants to buy an extra 12.5 % to allow for overlap.

She can only buy ribbon in full metres, costing £4.95 per metre.

How much will she spend on ribbon? (5 marks)

Use **π** = 3.14

Show your calculations and/or workings out here:

Step 1: Diameter = 80 mm × 2 = 160 mm; Circumference = 2 × 80 × 3.14 = 502.4 mm [M1]

Step 2: Ribbon for 15 cakes = 15 × 502.4 mm = 7536 mm [M1]

Step 3: Add 12.5% overlap = 7536 × 1.125 = 8478 mm = 8.478 metres [M1]

Step 4: Ribbon sold in full metres --> Must buy 9 metres [A1]

Total cost = 9 metres × £4.95 = £44.55 [A1]

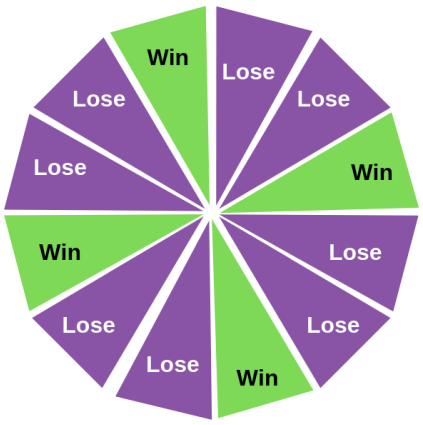
£44.55 (9 metres)

Write your answer in this box.

c) At the charity event there is a Wheel of Fortune game for the boys and girls.

To win you need to spin the dial and land on a 'win' segment.

Wheel of Fortune



15 girls and 15 boys are each having a turn on the game today.

What is the probability today that a child who plays is a girl, **and** that she wins a prize?
(3 marks)

Show your calculations and/or workings out here:

Step 1: Probability of player being a girl: 15 / (15 + 15) = 15/30 = 1/2

[M1]

Step 2: Probability of winning on Wheel: 4 win segments out of 12 = 4/12 = 1/3

Step 3: Combined independent probability = P(Girl) × P(Win):

P(Girl and Win) = 1/2 × 1/3 = 1/6 (or 0.167 / 16.7%) [M1 A1]

1 / 6 (or 16.7%)

Write your answer in this box.

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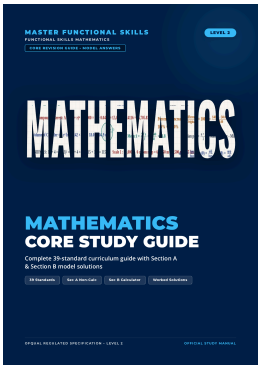

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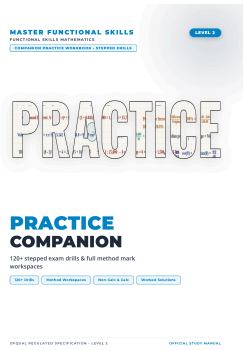

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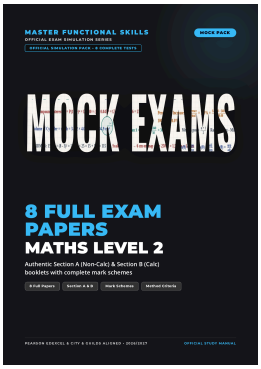

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